

Monocomponent Biosorption of Copper Ions (II) onto Nanocrystalline Cellulose from Coconut Husk Fibers

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Introduction: Nanocrystalline cellulose (NCC) is one of the most suitable cellulose derivatives for the treatment of wastewater. Various agricultural wastes have been used for the extraction of NCC. Coconut wastes have been widely studied as potential adsorbents for the removal of pollutants, including dyes and heavy metals.

Methods: In this work, nanocrystalline cellulose (NCC) was successfully isolated from coconut husk fibers through alkaline pretreatment accompanied by sulfuric acid hydrolysis. Then, the ability of NCC to adsorb Cu^{2+} from aqueous solution in batch studies was investigated.

Results: Results indicated that the optimal hydrolysis parameters were achieved at 50° C for 45 min with 64% sulfuric acid to extract NCC as rod-like particles with diameters between 4-10 nm. The potential of NCC as a biosorbent to remove copper ions (Cu^{2+}) from aqueous solution was investigated in terms of batch mode and maximum adsorption capacity (q_m) of 79.491 mg/g of Cu^{2+} . The adsorption efficiency of Cu^{2+} ions increased with an increase in the adsorbent dosage, decreased with an increase in the initial concentration of contaminant, and increased with the contact time. Under optimal conditions, adsorption kinetic followed a pseudo-second-order kinetic model and the adsorption isotherm fitted most closely with the Langmuir model.

Conclusion: According to a literature review, NCC from coconut husk fibers has not been used for the adsorption of heavy metals, mainly copper ions. This study shows that NCC from coconut husk fibers can be used as a low-cost and environmentally friendly adsorbent for the removal of Cu^{2+} from aqueous solutions.